collect_stats.py — technical

`scripts/collect_stats.py · 107 lines · scripts/collect_stats.py`

Purpose

EASY-EP statistics pass over the calibration set.

Step by step

Classes and functions in source order. Signatures, line ranges, raised exceptions and settings are extracted from the code; numbered steps describe what the code does.

`main`

```
def main() -> None
```

lines 29-103 · raises RuntimeError · settings CALIBRATION_BATCH_SIZE, MODEL_DTYPE

Script entry point: runs this stage end to end.

1. Load the original model and its task; discover all MoE layers; `prepare_calibration`.
2. Create an `ExpertStatisticsAccumulator`; the instrumentation callback checks outputs are finite, takes the layer's stack mask, intersects it with 'row is in the training split' for the bias sums, updates the accumulator and tracks the largest gate-consistency error.
3. For each batch (`CALIBRATION_BATCH_SIZE`): set the current masks/train rows, forward, accumulate the teacher-forced loss via `task.token_nll` (for logging).
4. Abort if the gate-consistency error exceeds the dtype tolerance.
5. `torch.save` the statistics; write the manifest (model, plugin, task, layer descriptions, calibration metadata and fingerprint, load report, loss, max gate error, runtime, memory, environment); log per-layer token counts.

Errors raised

Function	Exception	Message (f-string fields in braces)
`main`	RuntimeError	Gate consistency check failed during collection (rel. error {max_gate_err})